

Logarithms and Their Properties

ANSWER KEY



Evaluating logarithms

1 Evaluate each logarithm exactly:

a $\log_2 32 = 5$

b $\log_3 \frac{1}{9} = -2$

c $\log 1000 = 3$

d $\ln e^7 = 7$

2 Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

$$\log_5 18 = \frac{\ln 18}{\ln 5} \approx \frac{2.8904}{1.6094} \approx 1.796.$$

3 State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

Domain of f : $x > 4$. Inverse of g : $g^{-1}(x) = \log_2 x$.

Properties of logarithms

4 Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.
 $3\ln x + \frac{1}{2}\ln y - 2\ln z$.

5 Write as a single logarithm: $2\log x + 3\log y - \log z$.
 $\log\left(\frac{x^2y^3}{z}\right)$.

6 Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6 = 1.114$

b $\log_b 8 = 1.293$

c $\log_b 1.5 = 0.252$

Graphs of logarithmic functions

7 For $y = \log_2(x + 4) - 1$, state:

a the domain $x > -4$

b the vertical asymptote
 $x = -4$

c the x-intercept
 $0 = \log_2(x + 4) - 1 \Rightarrow x = -2$

8 Describe how the graph of $g(x) = -\ln(x - 2) + 3$ is obtained from $y = \ln x$.

Shift right 2 units, reflect across the x-axis, then shift up 3 units.

Logarithmic scales and applications

- 9 The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.
- a An earthquake has $M = 6$. Express I in terms of I_0 . $I = I_0 \cdot 10^6$.
- b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake? Difference in magnitude is 2, so intensity ratio is $10^2 = 100$ times.
- 10 Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?
 $\Delta\beta = 10\log(2) \approx 10(0.3010) \approx 3.01$ dB.
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More practice with logarithm properties

- 11 Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.
 $\log_2 8 + 2\log_2 x - \frac{1}{3}\log_2 y = 3 + 2\log_2 x - \frac{1}{3}\log_2 y$.
- 12 Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.
 $\ln(16^{1/2}) + \ln 3 - \ln 2 = \ln 4 + \ln 3 - \ln 2 = \ln\left(\frac{4 \cdot 3}{2}\right) = \ln 6$.